

Smart India Hackathon 2026: Official Problem Statement Details

Problem Statement Code: SIH-26138 • Project SagarQuantum (GreenFleet Quantum)

1. Problem Statement Identification & Metadata

Parameter	Official SIH Details
Problem Statement Code	SIH-26138
Project Title	Quantum-Inspired Multi-Objective Maritime Fleet Speed & Voyage Optimization for Decarbonization and Emission Reduction
Project Codename	Project SagarQuantum (GreenFleet Quantum)
Organization / Sponsor	Egreen Quanta
Category & Sector	Software & Clean-Tech Maritime Decarbonization, Quantum Computing & AI Logistics
Target Users	Fleet Operators, Ship Masters / Captains, Port Authorities, Maritime Regulators

2. Background & Problem Context Provided in SIH

- **Global Emission Burden:** International maritime shipping transports over 90% of global trade and produces ~3% of worldwide CO2 emissions (~1 billion tonnes/year), surpassing commercial aviation. Bunker fuel accounts for 50% to 60% of total voyage OPEX.
- **The Operational Flaw of Flat Speeds:** Ships conventionally maintain fixed cruise speeds (e.g. 18 knots), ignoring dynamic ocean currents and weather assistance. Under Admiralty's Law ($P \propto v^3$), sailing just 10% faster increases fuel consumption by 33%.
- **The 'Sail Fast, Wait at Anchor' Trap:** Vessels rush across oceans only to queue outside congested ports, burning fuel and incurring \$15,000 to \$40,000/day in port demurrage fines.
- **Regulatory Mandates:** The IMO Carbon Intensity Indicator (CII) grades ships from A to E (Grade E risks detention). The EU Emissions Trading System (EU ETS) levies carbon taxes of €75 to €90 per tonne of CO2e.

3. Core Challenge & Key Requirements

- **Multi-Objective Optimization:** Jointly minimize fuel burn, carbon taxes, demurrage, and CII penalties while guaranteeing strict on-time arrival.
- **Dynamic Environmental Ingestion:** Automatically ingest real-time 1/12-degree satellite ocean currents, wave height (Hs), wave period (Tp), and wind drag vectors.
- **Quantum Computational Acceleration:** Solve combinatorial speed search spaces ($>10^9$ choices) in real time (<150 ms) without getting trapped in local storm cost minima.
- **Regulatory Compliance:** Maintain IMO Grade-A compliance targets and generate verifiable, tamper-evident carbon audit certificates.
- **Fleet Swarm Coordination:** Enable multi-vessel Just-In-Time (JIT) port arrival synchronization to eliminate port bottleneck queues.

4. Expected Deliverables & Evaluation Criteria

Evaluation Pillar	What the Hackathon Evaluates	How GreenFleet Quantum Fulfills It
Optimization Performance	Quantifiable fuel & carbon reduction ($>10\%$).	14.0% to 18.2% verified fuel reduction across global trade corridors.
Solver Latency	Bridge-deployable real-time speed.	Sub-150 ms convergence via 3-Tier HQOA and 0.42 ms ML surrogates.
Real Environmental Data	Live integration with ocean & weather APIs.	Live Copernicus CMEMS (1/12° currents), AISStream, and OpenMeteo feeds.
Quantum Computing	Concrete quantum or quantum-inspired algorithms.	Real IBM Quantum 156Q Heron QAOA circuits with ZNE + QPSO tunneling.
System Usability	Functional dashboard for captains & fleet managers.	Industrial 13-screen React 19 / Leaflet GIS console with dark-mode UI.
Compliance & Verification	Verifiable, audit-ready carbon accounting.	Instant SHA-256 tamper-proof Grade-A certificates with PDF/XML exports.